Project Title

IoT-Based Laser Security and Intrusion Detection System Using ESP32 with Real-Time Web Monitoring

Submitted by:

Mayank Pal

24BEE0103

Objective

The objective of this project is to design and develop an IoT-enabled laser security system capable of detecting unauthorized intrusion using a laser beam and an LDR sensor. The system provides real-time visual and audible alerts through an RGB LED, buzzer, and LCD display while simultaneously hosting a web server on the ESP32 for remote monitoring of the security status over a Wi-Fi network.

Components Used

S.No	Component	Quantity	Purpose
1	ESP32-WROOM Development Board	1	Main controller with Wi-Fi
2	Laser Module (5V)	1	Continuous laser beam source
3	Digital LDR Sensor Module	1	Detects interruption of laser beam
4	16×2 LCD with I2C Module	1	Displays system status
5	RGB LED	1	Visual indication of system state
6	Active Buzzer	1	Audible intrusion alert
7	Breadboard	1	Circuit assembly
8	Jumper Wires	As required	Electrical connections
9	USB Cable	1	Programming and power supply
10	Wi-Fi Network	1	Web server connectivity

Simulation Images

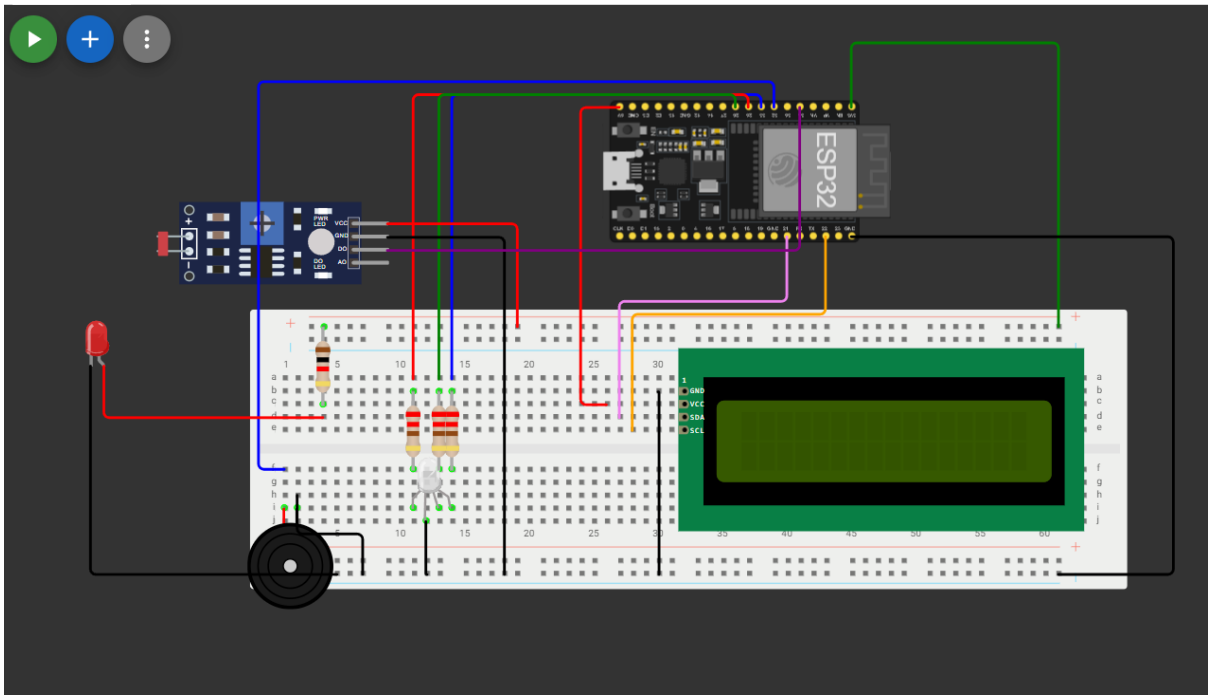


Figure 1: ESP32-Based Laser Security System Block Diagram

- System Block Diagram
- ESP32 Circuit Diagram
- Web Server Interface Screenshot
- Serial Monitor Output
- LCD Display during Secure Condition
- LCD Display during Intruder Alert

Hardware Images

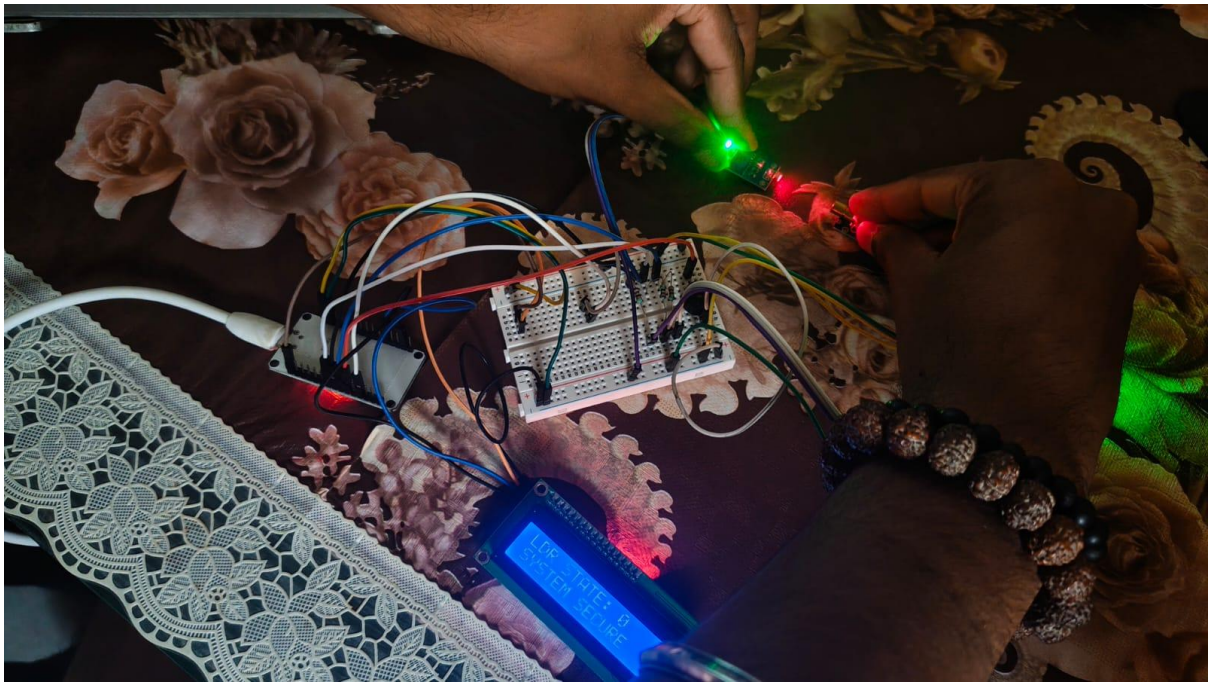


Figure 2: Final Hardware Prototype of the IoT Security System

Include:

- Complete Hardware Setup
- ESP32 Development Board
- Laser and LDR Alignment
- LCD Display
- RGB LED Status
- Web Server Running on Browser

Webserver Interface:

ESP32 LASER SECURITY SYSTEM

LDR State : 0

SYSTEM SECURE



ESP32 LASER SECURITY SYSTEM

LDR State : 1

INTRUDER ALERT

Circuit Connections

Component	ESP32 Pin
LDR Digital Output	GPIO34
LCD SDA	GPIO21
LCD SCL	GPIO22
RGB LED (Red)	GPIO25
RGB LED (Green)	GPIO26
RGB LED (Blue)	GPIO33
Buzzer	GPIO32
LCD VCC	5V
LCD GND	GND
LDR VCC	3.3V
LDR GND	GND
RGB LED Common Cathode	GND
Buzzer Negative	GND
Laser Module	External 5V Supply

Results and Observations

The developed laser security system successfully detected interruptions in the laser beam using the digital LDR sensor. During normal operation, the LCD displayed **"SYSTEM SECURE"**, the green LED remained ON, and the buzzer stayed OFF. When the laser beam was interrupted, the ESP32 immediately activated the red LED and buzzer while updating the LCD with **"INTRUDER ALERT"**. Simultaneously, the onboard web server displayed the current LDR status and security condition in real time, allowing wireless monitoring through any device connected to the same Wi-Fi network. The overall system demonstrated stable operation, fast response time, and reliable communication between hardware and web interface.

Future Scope

- Integration of ESP32-CAM for automatic image capture during intrusion.
- Cloud-based data logging using Firebase or ThingSpeak.
- Mobile application for remote monitoring.
- Instant SMS, Email, or Telegram notifications.
- AI-based face recognition for identifying authorized users.
- Battery backup for uninterrupted operation.
- Multi-zone security monitoring using multiple sensors.
- Motion sensor and PIR sensor integration for enhanced security

Conclusion

An IoT-based laser security system was successfully designed and implemented using the ESP32 microcontroller. The system effectively monitored the laser beam using a digital LDR sensor and generated immediate visual, audible, and LCD alerts whenever an intrusion occurred. The integrated web server enabled real-time wireless monitoring of the security status, demonstrating the practical application of IoT in modern security systems. The project achieved its objectives with reliable performance while maintaining a simple, cost-effective, and scalable architecture suitable for academic demonstrations as well as small-scale real-world security applications.

References

1. Espressif Systems, **ESP32 Series Datasheet**, Espressif Systems.
2. Espressif Systems, **ESP32 Arduino Core Documentation**.
3. Arduino Documentation, **Arduino IDE User Guide**.
4. LiquidCrystal_I2C Library Documentation.
5. ESP32 WebServer Library Documentation.
6. Random Nerd Tutorials, *ESP32 Web Server with Arduino IDE*.
7. IEEE Xplore Digital Library – Research papers on IoT-based Security Systems.
8. Manufacturer datasheets for the Digital LDR Module, RGB LED, and Active Buzzer.